Iteration 45

Unified University Calendar

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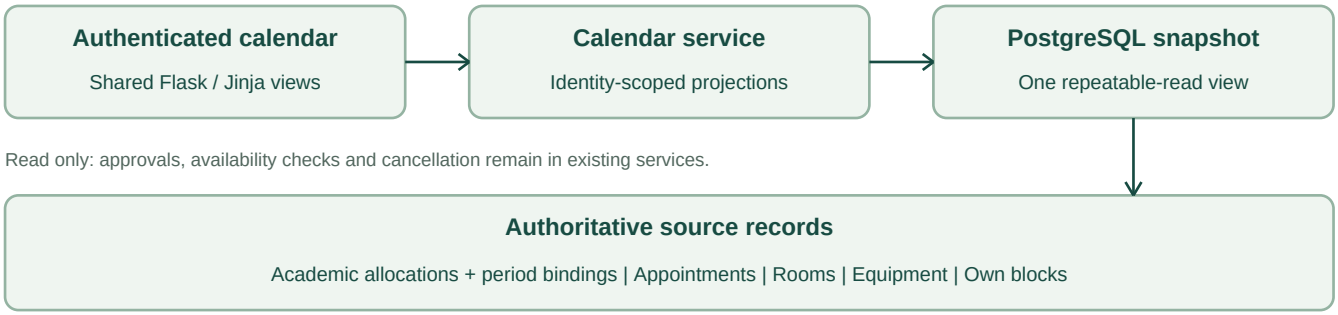
1. Objective and delivered experience

Bring university activities into one authenticated calendar without duplicating bookings or allocation logic. All roles can navigate month, week and day views, use Previous/Next/Today, filter categories and statuses, and open authorized event details. Mobile month views become a dated agenda; hourly views support horizontal and dense-cell scrolling.

2. Architecture and responsibilities



Component	Responsibility
UniversityCalendarService	Projects authoritative records into stable, display-only events; scopes SQL queries by identity, participation and current staff assignment.
Calendar blueprint / event macro / calendar.css	Authenticated navigation and filters, shared event cards, accessible status text, responsive views and safe error/empty states.
calendar_service / equipment_access	Reuse explicit bounded weekly expansion, timezone validation and current authorized custodian scope.
Existing domain services and EngineGateway	Keep decisions, reservations, physical custody, C++ scheduling and all JSON operations unchanged.

3. Source visibility and semantics

Source	Who sees it / what it means
Appointments	Organizer, required professors and invited students; revision approval progress is visible. Administrators have no unrelated private override.
Room activities	Organizer, invited professor and currently authorized assigned faculty reviewer; review work is not personal busy time.
Equipment	Borrower and current faculty custodian. Pickup/due/verified handover and return are point reminders, never participant-busy intervals.
Academic and personal	Known requesters/committee identities and own busy slots/blocks; explicitly published academic metadata is a separate audience, not enrollment.

4. Database and calendar integrity

Migration **0045a** adds two fields to resource_request: calendar_public (NOT NULL, default false) and nullable indexed academic_instructor_id (foreign key to university_user). A database check prevents publishing committee meetings. Administrator-only publication is CSRF protected and recorded in request history. There are still **47 application tables** plus Alembic infrastructure; no duplicate calendar-event table.

Europe/Istanbul is the configurable university timezone. All dated intervals are half-open [start, end). Weekly data expands only through explicit academic validity, planning weeks and exceptions. Missing legacy dates produce a warning, never invented commitments. Cancellations and rescheduling follow the source records; cancellation takes precedence over a previously declined optional invitation. Past completion is display-only and does not rewrite lifecycle state.

5. Executed verification

Check	Actual local result
Fresh C++ compilation and complete suite	172 passed, 0 failed; existing algorithms and protocols unchanged.
Complete Python / real PostgreSQL suite	314 run: 313 passed, 0 failed, 1 skipped on Windows (POSIX executable-permission test). Includes migrations, recovery, concurrency, JSON and five-strategy parity.
New calendar regressions	21 passed (included in the full total); additional final cancellation/filter follow-ups: 3 passed.
Additional Flask / CLI smoke	27 authenticated Flask page checks passed across student/professor/admin with runtime file access forbidden. Isolated CLI smoke: 2 passed.
Browser acceptance	Student month/week/day, navigation, pending/category filters and detail links; professor/staff scoping; 390px mobile layout; no console warnings/errors.
Migration and preservation	Fresh-schema and roundtrip checks passed; Alembic reports no drift at 0045a. Primary upgrade preserved all 47 preexisting table fingerprints; pending examples and generated CSV hashes unchanged.

Commands: python -m unittest discover -s tests -p test_*.py -v; fresh g++ -std=c++17 compilation of src (excluding main.cpp) plus tests; tests/test_backend_runtime.ps1; scripts/generate_database_diagrams.py --check. Tests use isolated PostgreSQL schemas, not the operational database. Remote CI accompanies the draft pull request; this report records local evidence.

6. Design decisions and remaining boundaries

Reuse application-service and read-projection boundaries rather than introducing another availability engine or calendar framework. Query-scoped privacy and a repeatable-read snapshot protect private data and prevent mixed-source views. Existing C++ Strategy/Factory components are retained without modification. Publication is explicit because no authoritative enrollment/publication model previously existed.

There is no inferred enrollment or automatic teaching assignment. An instructor label is presentation metadata, not a new personal commitment. Only the active academic allocation run is projected. Views refresh on navigation/reload rather than live push, and fail clearly above 5,000 entries. Legacy dates still require explicit configuration. No external calendar synchronization, new scheduling feature, deployment or strategy benchmarking is included.

7. Delivery artifacts

Architecture: docs/architecture/unified-calendar-design.md. Current requirements: docs/srs/unified-calendar-requirements.md. Database Mermaid/SVG diagrams and catalog match the schema. README, database operations and demonstration walkthrough describe the calendar. Delivery uses iteration/45-unified-university-calendar and a draft PR; final merge remains subject to review.